1 S-R-A LOOP

S-R-A loop is repeating process where:

1. Seasoning,
2. Reasoning and
3. Acting

is involved during the procedure of controlling the robot. This is the most important part of software in robotics. Remember the `autonomous` control is `ability to perform intended tasks based on current state and sensing, without human intervention`.

1.1 Tasks:

1. Write a program to drive the robot around the class and avoid the obstacles.
2. Using the S-R-A loop technique you should write the program in particular order:
    1. Check the sensor. IF the bumper ...
    2. ... Is pressed the robot has to stop/go back/turn.
    3. ... Is not pressed the robot can drive forward.

Program 1: Detecting the obstacle.

```
const int BUMPER_PIN = A0;
void setIOpins();
void robotForward();
void robotStop();
void robotLeft();
void robotRight();
void robotBackward();

void loop() {
  bool bumperIsPressed = digitalRead(BUMPER_PIN);
 if ( bumperIsPressed ){
 robotStop();
 }else{
 robotForward();
 }
}
[+]void setup() { ...
```

1.2 Questions:

1. What for `S-A-R loop` stands for?

2. Mark all three basic S-A-R processes in previous code example.
3. Can `line 4` of the previous code example be written outside of `loop()` function?

1.3 Summary:

1.3.1 <++>

<++>

1.4 Issues:

1.4.1 <++>

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